

Exercise: Updating SQL

Context

You are working with a product inventory system that tracks products, categories, and pricing. Your job is to clean and update data as part of a system maintenance task.

Environment

1. Create a new PostgreSQL database named `inventorydb`.
2. Run all setup scripts in this exercise against the `inventorydb` database.

Setup Script (Run first)

SQL

```
CREATE TABLE ProductCategory (  
    CategoryId INT PRIMARY KEY,  
    CategoryName VARCHAR(50)  
);  
  
CREATE TABLE Product (  
    ProductId INT PRIMARY KEY,  
    ProductName VARCHAR(100),  
    CategoryId INT,  
    Price DECIMAL(10, 2),  
    EndDate DATE,  
    FOREIGN KEY (CategoryId) REFERENCES ProductCategory(CategoryId)  
);  
  
INSERT INTO ProductCategory VALUES  
(1, 'Books'),  
(2, 'Stationery'),  
(3, 'Clearance');  
  
INSERT INTO Product VALUES  
(101, 'Notebook', 2, 5.00, NULL),  
(102, 'Pen Set', 2, 7.50, NULL),  
(103, 'Calendar 2023', 2, 9.99, NULL),  
(104, 'Mystery Novel', 1, 15.00, NULL),  
(105, 'Classic Fiction', 1, 12.50, NULL);
```

Instructions

Part 1: Update a Single Record

1. Update the price of 'Notebook' to \$6.25.

Part 2: Update Multiple Columns

2. Update 'Pen Set' to:
 - New name: 'Executive Pen Set'
 - New price: \$8.99

Part 3: Update Multiple Records

3. Set the EndDate to '2024-12-31' for all products in the 'Stationery' category.

Hint: Use a subquery or CategoryId directly based on the data provided.

Part 4: Handle Foreign Key Relationships

4. Try to update the CategoryId of 'Calendar 2023' to a value that does not exist (e.g., 999).
 - What happens? Record the error message.
 - Then, update it to 'Clearance' (CategoryId 3) instead.

Part 5: Bonus Challenge

5. Write a single UPDATE statement to change 'Classic Fiction' to:
 - New name: 'Vintage Novel'
 - New category: 'Clearance'
 - New price: \$10.00

Reflection Questions

- Why is it important to run a SELECT query before updating?
- What risks are associated with forgetting a WHERE clause?
- How do foreign keys protect data consistency during updates?